

Xinyi Guan

(+1) 312-810-3169 | guanxinyi0916@gmail.com | linkedin.com/in/xinyi-guan-42016928b | github.com/Xinyidududu

EDUCATION

Purdue University

Bachelor of Science in Computer Science, Concentrate on Software Engineering — **GPA: 3.75/4.0**

West Lafayette, IN

January 2022 - May 2026

Columbia University

Master of Science in Artificial Intelligence, Concentrate on AI Infrastructure

New York, NY

Expected: December 2027

TECHNICAL SKILLS

Program Languages: Java, C, C#, C++, Rust, R, Python, HTML/CSS, JavaScript, Visual Basic.NET, SQL, Dart, Swift

Developer Tools and Frameworks: Git, Google Cloud Platform, PuTTY, CVAT, Jupyter Note, Vocareum, Coverity, CodeQL, XCode, Android Studio, Facebook Infer, MySQL, SQLite, NoSQL, Microsoft Office Suite, AWS, JWT, Flask, Spring Boot, Flutter, Unity, JUnit, Tor Network, Node.js, MongoDB, Postgresql, Docker

Languages: English, Mandarin

INTERNSHIP

ARC Rentals LLC

Software Engineer Intern

May – August 2025

Remote

- Collaborated with a cross-functional team to design and develop a rental application for Veygo, a car rental application.
- Built key front-end features using **Swift** in **Xcode**, integrating seamlessly with a **Rust**-based backend and **PostgreSQL** database.
- Implemented an on-device AI feature leveraging **Apple's Foundation Model** to generate structured city insights dynamically rendered in SwiftUI.

RESEARCH

The Data Mine Program Researcher

Assessing Reactive Groups and Alerts in the CAMEO Chemicals Database

January - May 2024

West Lafayette, IN

- * Conducted data mining and analysis on the CAMEO Chemicals Database, in collaboration with CAS, to explore the relationship between reactive chemical groups and associated reactivity alerts.
- * Implemented **Python-based** data analysis techniques, including **K-means clustering** and **the elbow method**, to identify patterns and correlations in the dataset.
- * Utilized **cross-correlation heatmaps** and **PCA plots** to present findings and aid in the understanding of chemical reactivity patterns.

Vertically Integrated Project Researcher

Analysis of Food Processing Crystal Image

August - December 2023

West Lafayette, IN

- * Improved manufacturing processes and product quality by developing an automated analysis tool for sugar crystal microscope images with TinyML teammates and Harvard researchers.
- * Used **Python** to implement a machine learning model based on **Hough Transform** for bubble detection and removal.
- * Optimized image pre-processing techniques, including noise removal, binarization, and adaptive thresholding, and integrated the **Stardist model** for precise crystal separation.

PROJECTS

Bareturn0; | Unity, AWS, MongoDB, Node.js, JavaScript, C#, Git, Figma

January - May 2025

- * Developed a full-stack educational card game with card game mechanics that integrates programming concepts and basic math into gameplay, enhancing learning through interactive storytelling and gamification with a team of 4.
- * Engineered a secure backend with **Node.js** and **JavaScript**, and develop the frontend with **C#** and **Unity**. Used **MongoDB** as database, **AWS** as cloud platform. Created and designed an intuitive user interface and user flow with **Figma**.

FooMyFood | Spring Boot, Flutter, MySQL, Google Cloud, Java, Dart, Git

August - December 2024

- * Designed a mobile app to help individuals track food inventory, receive expiration reminders, and get personalized recipe suggestions, reducing food waste and simplifying meal planning with a team of 4.
- * Developed a full-stack application using **Dart** and **Flutter** for the frontend, **Spring Boot** and **Java** for the backend, **MySQL** for database management and used **Google Cloud** as our cloud platform. Created an intuitive user interface and user flow with **Figma**, enhancing accessibility and engagement.

- * Developed a Python-based secure publish-subscribe blog application using the **Bento** framework, integrated with the **Flask** web framework and **SQLite** database.
- * Enhanced user anonymity by incorporating the **Tor network**, enabling access through a .onion domain.
- * Implemented key functionalities including user registration, authentication with **JSON Web Tokens (JWT)**, message publishing, deletion, and recovery, utilizing Bento's API for secure message handling.